



Exponentials and Logarithms · Worksheet 1

Chapter 6: Exponentials and Logarithms. Answer all questions. Show your working clearly. Total: 105 marks. Time: 2 hours 10 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are five sections, one for each topic in Chapter 6. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- In every logarithmic equation, **check each solution** against the domain of the original logs.
- Give non-exact answers to 3 significant figures unless stated otherwise.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Laws of logarithms	18	
B Solving exponential and logarithmic equations	20	
C Graphs of a^x , e^x and $\ln x$	19	
D Exponential growth and decay models	22	
E Reducing a relationship to linear form	22	
Roar-level challenge	4	
Total	105	

Section A Laws of logarithms

18 marks

1. Write $2\log_2 3 + \log_2 5 - \log_2 15$ as a single logarithm in its simplest form.

[2]

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2. Find the exact value of (a) $\log_4 8$, (b) $\log_3 \left(\frac{1}{27} \right)$. [2]

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3. Given that $p = \log_a x$, $q = \log_a y$ and $r = \log_a z$, express [3]

$$\log_a \left(\frac{x^2 y^3}{\sqrt{z}} \right)$$

in terms of p , q and r .

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4. Solve $\log_6 x + \log_6 (x + 5) = 2$. [4]

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5. (a) Given that $\log_2 y = 3 + 2\log_2 x$, express y in terms of x , giving your answer in a form not involving logarithms. [3]

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- (b) Solve [4]

$$2\log_3 x - \log_3(x - 2) = 2.$$

Show clearly that each of your solutions is valid.

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(Total for Question 5 is 7 marks)



Section B Solving exponential and logarithmic equations

20 marks

6. Solve $3^x = 20$, giving your answer to 3 significant figures.

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7. Solve $2^{x+1} = 5^x$, giving your answer to 3 significant figures.

[3]

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8. Solve $\ln(2x + 3) - \ln x = \ln 5$.

[3]

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9. By using the substitution $y = 2^x$, solve

[4]

$$2^{2x} - 6 \times 2^x + 8 = 0.$$

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10. (a) Solve $e^{2x} - 5e^x + 6 = 0$, giving your answers as exact values.

[4]

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(b) Show that the solution of

[4]

$$2^x \times 3^{2x+1} = 15$$

can be written as $x = \frac{\ln a}{\ln b}$, where a and b are integers to be found.

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(Total for Question 10 is 8 marks)

**Section C** Graphs of a^x , e^x and $\ln x$

19 marks

11. For the curve $y = e^x + 2$, write down [2]

- (a) the coordinates of the point where the curve meets the y -axis,
- (b) the equation of the asymptote.

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12. For the curve $y = \ln(x + 4)$, state [3]

- (a) the domain,
- (b) the equation of the asymptote,
- (c) the coordinates of the point where the curve meets the x -axis.

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13. Sketch the curve $y = 2e^x - 5$. [3]

Show the exact coordinates of the points where the curve meets the axes, and the equation of the asymptote.



14. The curves $y = e^{2x}$ and $y = 4e^{-x}$ intersect at the point P .
Find the exact coordinates of P .

[4]



15. The functions f and g are defined by

$$f(x) = \ln(2x - 3), \quad x > \frac{3}{2}, \quad g(x) = e^x, \quad x \in \mathbb{R}.$$

- (a) Sketch the graph of $y = f(x)$, showing the equation of the asymptote and the coordinates of the point where the graph meets the x -axis. [2]

- (b) Find $f^{-1}(x)$ and state its domain. [3]

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- (c) Solve $fg(x) = \ln 5$, giving your answer in the form $k \ln 2$. [2]

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(Total for Question 15 is 7 marks)



Section D Exponential growth and decay models

22 marks

16. The number of bacteria, N , in a culture is modelled by $N = 500e^{0.2t}$, where t is the time in hours. [2]

Find the number of bacteria (a) at the start, (b) after 5 hours.

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17. The value, $\pounds V$, of a car t years after it was bought is modelled by $V = 20\,000e^{-0.15t}$. [3]

(a) Find the value of the car after 4 years.

(b) Find the year in which the value of the car first falls below $\pounds 5000$.

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18. The mass, M grams, of a radioactive substance after t days is modelled by $M = M_0e^{-kt}$. [4]
The half-life of the substance is 8 days.

(a) Show that $k = \frac{\ln 2}{8}$.

(b) Find the percentage of the original mass remaining after 30 days.

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19. The population, P , of a town t years after 2020 is modelled by $P = Ae^{kt}$. [4]
In 2020 the population was 2000, and in 2023 it was 3500.
- (a) Find the value of A and the value of k .
- (b) Use the model to predict the population in 2030.

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20. A cup of tea is left to cool in a room. Its temperature, $T^{\circ}\text{C}$, t minutes after it is made is modelled by

$$T = 20 + 75e^{-kt},$$

where k is a positive constant.

(a) State the initial temperature of the tea. [1]

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(b) After 5 minutes the temperature of the tea is 60°C . [3]

Show that $k = \frac{1}{5} \ln \left(\frac{15}{8} \right)$.

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(c) Find the time taken for the temperature of the tea to fall to 30°C . [3]

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(d) State the value that T approaches as t increases, and give one limitation of the model. [2]

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(Total for Question 20 is 9 marks)



Section E Reducing a relationship to linear form

22 marks

21. Two variables are related by $y = ax^n$. A graph of $\log_{10} y$ against $\log_{10} x$ is a straight line with gradient 3 and intercept $\log_{10} 5$ on the vertical axis. Find a and n . [3]

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22. Two variables are related by $y = ab^x$. A graph of $\ln y$ against x is a straight line with gradient 0.4 which passes through the point $(0, 1.2)$. Find the values of a and b to 3 significant figures. [3]

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23. Show that $y = 4 \times 3^x$ can be written in the form $\log_{10} y = \log_{10} 4 + x \log_{10} 3$. Hence state the gradient and the vertical intercept of the graph of $\log_{10} y$ against x . [3]

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24. The table shows some experimental data which are thought to follow a relationship of the form $y = ax^n$. [4]

	Point 1	Point 2
$\log_{10} x$	0.301	0.699
$\log_{10} y$	1.083	1.877

Both points lie on the line of best fit of $\log_{10} y$ against $\log_{10} x$.

Find the values of n and a , giving each to 1 significant figure, and hence write down the relationship between y and x .

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25. The value, £ V , of a painting t years after the year 2000 is modelled by

$$V = pq^t,$$

where p and q are constants.

- (a) Show that $\log_{10} V = \log_{10} p + t \log_{10} q$, and explain why this means that a graph of $\log_{10} V$ against t is a straight line. [2]

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- (b) The graph of $\log_{10} V$ against t passes through the points $(0, 3.2)$ and $(10, 3.9)$. [3]
Find the values of p and q , giving your answers to 3 significant figures.

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- (c) Interpret the value of q in the context of the model. [2]

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- (d) Use the model to predict the value of the painting in 2025, to 2 significant figures. [2]

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(Total for Question 25 is 9 marks)



ROAR-LEVEL CHALLENGE

4 marks

26. Solve the equation $\log_2 x = \log_4(x + 6)$.

[4]

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END OF WORKSHEET

Total: 105 marks